

Transcription as a source of genome instability

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Abstract | Alterations in genome sequence and structure contribute to somatic disease, affect the fitness of subsequent generations and drive evolutionary processes. The crucial roles of highly accurate replication and efficient repair in maintaining overall genome integrity are well-known, but the more localized stability costs that are associated with transcribing DNA into RNA molecules are less appreciated. Here we review the diverse ways in which the essential process of transcription alters the underlying DNA template and thereby modifies the genetic landscape.

Changes to genomic DNA can be in the form of mutations that alter the primary sequence or rearrangements that alter chromosome structure. Most mutations and rearrangements are assumed to arise either randomly during the process of genome duplication or in response to DNA damage, and multiple repair pathways have evolved to maintain such changes at an acceptably low level. The consequences of too much genetic instability are particularly evident in humans, where a loss of repair capacity is associated with cancer predisposition syndromes and with ageing. Whereas replication involves making precisely one copy of each DNA strand throughout the genome, the transcription of DNA into RNA products is a comparatively non-uniform process; it only affects defined segments of the genome, it typically copies only one strand of the DNA, and it occurs at highly variable rates.

The first indication that transcription might do more than just passively copy the DNA template came from bacterial studies in the 1970s demonstrating that exogenous mutagens were more efficient at inducing mutations if the reporter gene was highly transcribed^{1,2}. It was not until 15 years later that transcription was also demonstrated to stimulate spontaneous mutagenesis in eukaryotes, specifically in budding yeast³. This phenomenon, which locally and permanently alters the primary sequence of the DNA template, is referred to as transcription-associated mutagenesis (TAM). It should be noted that TAM is not to be confused with so-called 'transcriptional mutagenesis', which refers to the production of mutant mRNAs and proteins through transient alterations in the DNA template⁴. The discovery of transcription-associated recombination (TAR) came through identification of *HOT1* in yeast; this is

a sequence that stimulates mitotic recombination by promoting high levels of transcription^{5,6}. Transcription thus has the potential to modify the genetic landscape by locally altering mutation rates, by stimulating loss of heterozygosity and by generating diverse types of rearrangements that include deletions, duplications, inversions and translocations.

Studies of TAM and TAR typically require a selective system to identify rare mutants and recombinants, respectively, as well as regulation of the system by a promoter with activity that can be tightly controlled. Depending on the system used, the difference between low and high transcription levels can be orders of magnitude. Microorganisms such as *Escherichia coli* and *Saccharomyces cerevisiae* have been the experimental organisms of choice for practical reasons: rapid growth, ease of genetic manipulation, well-defined replication origins and availability of reporter systems. In general, specific mechanisms of TAM and TAR have been deduced by studying the effects of depleting individual proteins that are involved in DNA and/or RNA metabolism. This is trivial to do through standard gene-targeting techniques in microorganisms and now analogous studies in metazoans have become more feasible through the development of RNAi technologies.

Recent efforts have moved away from simple descriptive analyses of TAM and TAR and have focused on understanding how transcription destabilizes the underlying DNA template. The level of transcription is clearly important; in microorganisms, the rate of TAM is directly proportional to the transcription rate^{7,8}, and a similar proportionality has been reported for TAR in mammalian cells⁹. An early conceptual link between TAM and TAR was derived from the fact that mutation

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Non-transcribed strand (NTS). The NTS is complementary to the transcribed strand and has the same sequence as the RNA (except that it contains thymine instead of uracil); it is often referred to as the coding strand, whose sequence is given as standard.

Replisome

The multi-protein complex that contains all of the proteins that are required for DNA replication. This includes the DNA polymerases, factors that increase the processivity of DNA synthesis and a helicase to unwind duplex DNA.

Two-dimensional gels

These are used to visualize replication fork progression across a defined segment of DNA. DNA is separated by size in the first dimension and by shape in the second; the fragment of interest is visualized by Southern blot analysis. Linear fragments run on a diagonal; fragments that run off the diagonal correspond to replicating or branched molecules.

and recombination each reflect a normal mechanism for dealing with DNA damage. The initial assumption was that there might be a single source of the transcription-associated damage underlying TAM and/or TAR. Indeed, early studies indicated that TAM primarily reflects damage to the non-transcribed strand (NTS) of the DNA template, whereas TAR is largely due to transcription–replication conflicts. However, recent studies have made it apparent that there are multiple mechanisms that contribute to TAM and TAR. Importantly, transcription can affect stability of the template by mechanisms that are separate from DNA replication, potentially making transcription-associated alterations a key contributor to genetic changes in non-dividing cells.

In the following sections, we begin by considering the importance of replication–transcription conflicts to genetic instability and how persistent association of the RNA transcript with the template DNA strand exacerbates these conflicts. Next, we summarize recent studies that highlight how the primary sequence of actively transcribed DNA — in particular, its propensity to form non-B secondary structures — contributes to instability. Finally, our current knowledge of factors that contribute to TAM will be summarized. An important point to be borne in mind throughout is that even though an experimental observation or mechanism may currently be limited to a single organism, the high evolutionary conservation of DNA structure and of basic DNA metabolic processes makes it likely that documented mechanisms of TAM and TAR will be broadly applicable.

Transcription and replication collisions

Transcription and replication occur on the same template, making conflicts between these two processes inevitable. Whereas replication copies both strands of duplex DNA, only one strand is typically transcribed by RNA polymerase (RNAP); the transcribed strand can thus correspond to either the leading- or lagging-strand template of replication (FIG. 1). When the leading strand of replication is the transcribed strand, the replication and transcription forks move in the same direction; when the lagging strand is the template for transcription, the forks converge. We will refer to the resulting RNAP–replisome conflicts as co-directional collisions and head-on collisions, respectively. In addition to direct conflicts between transcription and replication, positive supercoils accumulate ahead of both machineries, thereby posing an additional problem for head-on collisions. Such unresolved helical stress can trigger replication fork reversal¹⁰, giving rise to a ‘chickenfoot’ structure that can be enzymatically processed into a recombination-initiating double-strand break¹¹.

The cost of head-on collisions. The relevance of RNAP–replisome collisions to genome instability has been extensively reviewed^{11–13}, and the general consensus is that head-on collisions are more destabilizing because they impede replication to a greater extent than co-directional collisions. An early indication that head-on collisions are more problematic came from the striking

co-directional orientation of all seven of the highly expressed ribosomal RNA (rRNA) operons in the *E. coli* genome¹⁴ — a pattern that has been observed in over 80 bacterial species¹⁵. Importantly, it was recently demonstrated that reversing the orientation of rRNA genes in *Bacillus subtilis* negatively affects fitness^{16,17}. In budding yeast, rRNA transcription units are insulated from head-on collisions by replication fork barriers, which physically block forks from entering the distal end of transcription units¹⁸. Finally, within 50 kb of putative human replication origins, it has been estimated that genes that are co-directionally transcribed with replisome movement outnumber those with a head-on orientation by eightfold¹⁹. Although genome instability can result from transcription-associated disruptions in replication, it should be noted that the disruption of transcription caused by replication can also be potentially costly for fitness.

What happens when a replication fork approaches an actively transcribed gene? At least in *in vitro* studies with purified *E. coli* components, a replisome moving in either direction appears to dissociate RNAP from the template as it passes¹². As for the effect on replication, analysis of forks using two-dimensional gels has detected discrete replisome pausing with the head-on RNAP–replisome orientation, but not with the co-directional orientation, in both *E. coli*²⁰ and yeast^{21–23}. More recently,

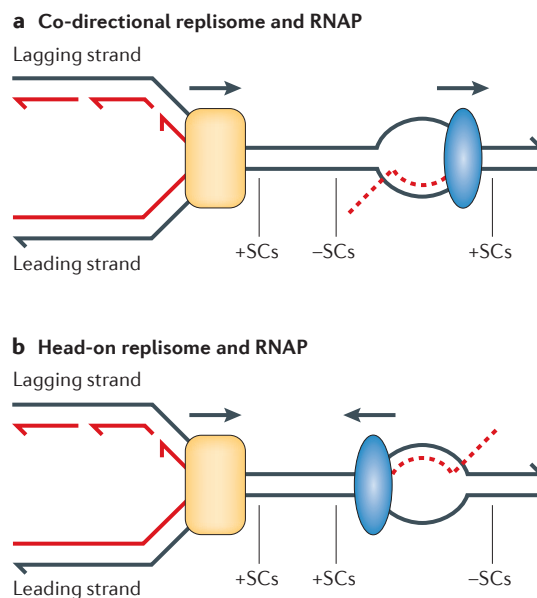


Figure 1 | Co-directional and head-on orientations of RNA polymerase and the replisome. **a** | In the co-directional orientation, the transcribed strand is the leading-strand template for replication. **b** | In the head-on orientation, the transcribed strand is the lagging-strand template. Positive supercoils (+SCs) accumulate ahead of replisome; +SCs and negative supercoils (–SCs) accumulate ahead of and behind the RNA polymerase (RNAP), respectively. Nascent DNA and RNA are indicated as solid and dashed red lines, respectively; arrowheads indicate the 3' ends of DNA strands. RNA polymerase is represented in blue, and the replisome is represented in yellow.

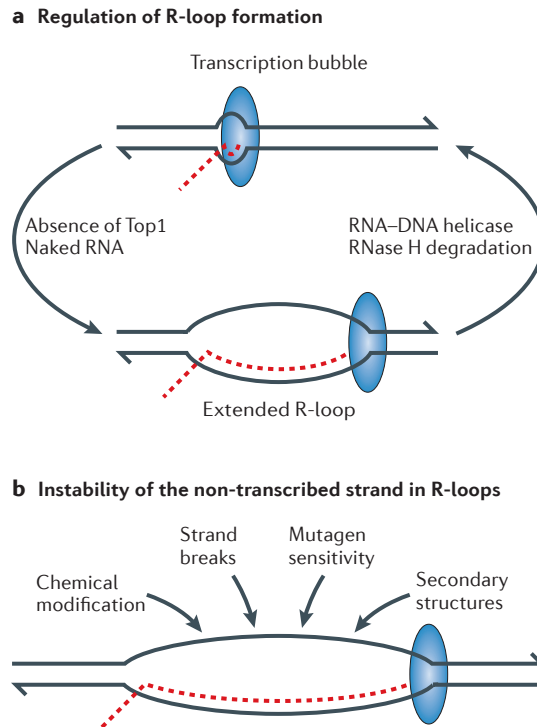


Figure 2 | Factors that promote and remove R-loops during transcription. **a** | R-loop formation is favoured by the negative supercoils that accumulate in the absence of topoisomerase 1 (Top1) and by naked RNA that fails to be engaged immediately after transcription. In bacteria, the coupling between transcription and translation prevents the accumulation of naked RNA. In eukaryotes, RNA is co-transcriptionally assembled into ribonucleoprotein particles for splicing and/or nuclear transport. R-loops can be actively unwound by an RNA-DNA helicase, or the RNA component can be degraded by RNase H. **b** | Factors expected to affect the exposed non-transcribed strand within R-loops. DNA strands are in black, and 3' ends are indicated by the half-arrowheads. The RNA transcript is shown by the red dashed line, and the large blue oval corresponds to RNA polymerase.

chromatin immunoprecipitation followed by microarray (ChIP-chip) analysis of sequences that are preferentially associated with a replicative DNA polymerase from yeast was used to infer the positions of slow fork movement throughout the genome²⁴. These sites were positively correlated with transcription, although in this case, a replication fork slowing was evident regardless of the relative orientation of transcription and replication. The two-dimensional gel and ChIP-chip results, however, should not be considered to be contradictory, as the resolution of pausing is very different in the two types of assays.

TAR as a replication-dependent event. It is well known that replication roadblocks can be bypassed, and replication at collapsed or broken forks can be re-established through homologous recombination²⁵. It is, therefore, not surprising that TAR has been linked to DNA replication. In yeast, for example, cell-cycle-regulated

promoters were used to show that only transcription in S phase was able to stimulate recombination²². A similar S phase connection has been inferred in mammalian studies, in which TAR was detected only in cycling cells²⁶. In the case of yeast, the recombination-promoting potential of co-directional versus head-on collisions between the replisome and RNAP has been specifically examined. Consistent with a greater impairment of replication fork movement by head-on collisions, TAR was enhanced in the head-on orientation relative to the co-directional orientation^{22,23,27}. A similar orientation-specific effect has been reported in an analysis of transcription-associated gross chromosomal rearrangements in yeast²⁸ and with transcription-associated deletions in a plasmid-based *E. coli* assay²⁹. Despite multiple examples of greater instability with head-on rather than with co-directional replisome-RNAP collisions, it should be noted that the relative instability reverses in *E. coli* when DNA polymerase encounters a stalled rather than a processive RNAP elongation complex³⁰.

An additional point to consider with respect to TAR concerns the recent demonstration that highly expressed yeast genes are tethered to the nuclear pore — a phenomenon that is referred to as ‘gene gating’³¹. This association facilitates mRNA export but creates an orientation-independent topological barrier that hinders replication fork progression³². One function of the replication checkpoint is to sever this connection, preventing topologically driven fork reversal and allowing replication to continue. Whether such gene gating is relevant to TAR has not been specifically examined.

Is replication fork direction relevant to TAM? In addition to stimulating recombination, an interesting possibility is that the transcription-associated slowing or pausing of replication forks might also lead to recruitment of translesion synthesis DNA polymerases. These polymerases are specialized to bypass lesions that stall replicative DNA polymerases and can have extraordinarily low fidelity on undamaged DNA templates³³. Their inappropriate recruitment would thus be expected to be mutagenic and could be a potential source of TAM. Although there is as of yet no compelling connection between replication fork direction and TAM in eukaryotes, a higher rate of mutagenesis with head-on versus co-directional RNAP-replisome movement has been reported in *B. subtilis*¹⁷. It is also intriguing to note that reversing the direction of replication through a highly expressed mutation reporter subtly affects the spectrum of TAM in yeast⁷.

Transcription-associated R-loops

RNA exits through a channel in RNAP as it is synthesized, thereby disrupting its base pairing with the complementary DNA template strand. However, the nascent RNA has the potential to anneal back to the transcribed strand, creating a stable RNA-DNA hybrid and leaving the NTS exposed as an extended single strand of DNA^{34–36}. This structure, which is called an R-loop, can be over 1 kb in length in highly transcribed genes (FIG. 2).

Chromatin immunoprecipitation followed by microarray (ChIP-chip). DNA that interacts with a given protein is immunoprecipitated from cell extracts (‘ChIP’). The precipitated DNA is labelled and hybridized to a microarray (‘chip’), where signals above background reflect sequences preferentially immunoprecipitated with the protein of interest; it is used to map locations of protein-DNA interactions.

Box 1 | Roles of Top1 in TAR and TAM

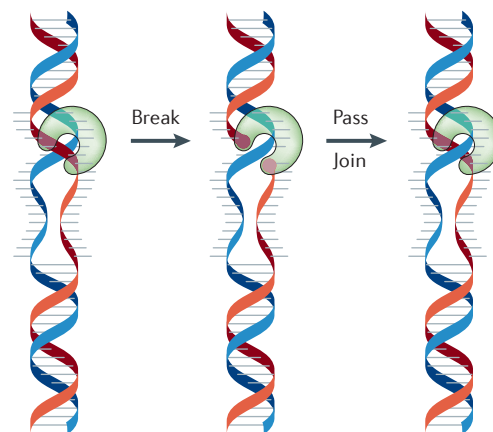
Topoisomerase 1 (Top1) has roles in transcription-associated recombination (TAR) and transcription-associated mutagenesis (TAM). Top1 relaxes the superhelical stress generated when DNA strands are unwound during transcription (figure, left). Top1 nicks one strand of the DNA (figure, centre), forming a covalent linkage with one end. The intact strand passes through the nick, and then Top1 reseals the nick (figure, right).

Top1 activity constrains TAR

In the absence of Top1, negative supercoils accumulate behind RNA polymerase (RNAP), giving the DNA single-stranded character and promoting the formation of R-loops. Top1 also regulates the activity of RNA splicing factors and ribonucleoprotein (RNP) assembly, thereby preventing the accumulation of the naked RNA that is required for R-loop formation. R-loops are a primary cause of TAR, presumably through their interference with replication fork progression.

Top1 activity promotes TAM

Top1 can become trapped during its cleavage–ligation cycle, giving rise to a nick with the enzyme that is covalently attached to one end. This irreversible complex is likely to be processed into a small gap. If the gap is within a short tandem repeat, misalignment between the complementary strands can bring the ends together to facilitate ligation and to create a deletion intermediate⁸³. If a ribonucleotide monophosphate (rNMP) is present at the Top1 cleavage site, the 2'-OH of ribose attacks the covalent Top1–DNA linkage to release Top1 completely (REF. 122). Although rNMP-dependent deletions might occur by a similar mechanism to that proposed for a trapped Top1 cleavage complex, an alternative possibility is that strand rejoining occurs by means of a second Top1 cleavage–ligation reaction¹²³.



Preventing the accumulation of R-loops. In bacteria, mRNA is immediately engaged in translation by ribosomes as it exits RNAP. When transcription and translation become uncoupled, transcription rapidly terminates, thereby blocking the expression of all downstream genes in polycistronic operons. This phenomenon, which is known as ‘polarity’, not only prevents the energy costs that are associated with the continued production of untranslated mRNA, it also blocks the accumulation of naked RNA that can potentially anneal back to the template DNA strand to form an R-loop³⁶. In eukaryotes, wherein transcription and translation occur in separate cellular compartments, nascent RNAs are co-transcriptionally assembled into ribonucleoprotein (RNP) particles that promote splicing and/or nuclear export. Although these co-transcriptional processes undoubtedly increase the efficiency of mRNA processing and/or transport, they also prevent the accumulation of naked RNA, as in bacteria. Even in those cases in which the RNA is the final gene product (for example, in rRNA and tRNA), either extensive formation of RNA secondary structure or rapid association with proteins discourages re-annealing of the transcript to the DNA template.

In addition to co-transcriptional RNA engagement, R-loops are also held in check by the activity of topoisomerase 1 (Top1) (BOX 1), which is an enzyme that relaxes superhelical stress in duplex DNA. During transcription, positive and negative supercoils accumulate ahead of and behind RNAP complexes, respectively, forming ‘twin domains’ of helical stress³⁷ (FIG. 1). Positive supercoils can impede further DNA unwinding, whereas excessive negative supercoiling imparts single-stranded character to duplex DNA and thereby promotes R-loop formation. In murine cells, TOP1 also

limits R-loop formation through the regulation of RNA splicing and RNP assembly factors³⁸.

If the mechanisms that normally prevent R-loop formation fail, there are back-up mechanisms in place to remove these structures. This is a major function of the RNase H class of enzymes, which specifically degrades the RNA component of RNA–DNA hybrids³⁹. The growth defect that is associated with a TOP1 deficiency, for example, can be rescued by overexpression of RNase H in bacterial or human cells^{38,40}. Finally, the Sen1 helicase of budding yeast and its human homologue senataxin (SETX), as well as the RecG helicase of *E. coli*, have been implicated in unwinding R-loop structures^{41–43}.

R-loops and genetic instability. The connection between R-loops and TAR initially emerged through study of the *hpr1* hyper-recombination mutant of yeast. Transcript elongation was shown to be impaired in *hpr1* mutants⁴⁴, and this defect, together with the hyper-recombination phenotype, was suppressed by RNase H overproduction⁴⁵. Hpr1 is a subunit of the THO complex, which interacts with the transcription–export (TREX) complex to facilitate mRNA export⁴⁶. A general model that emerged from these studies is that, in the absence of THO–TREX complex components, the nascent RNA fails to assemble properly into RNPs. This, in turn, promotes R-loop formation (which impedes transcript elongation), causes conflicts with replication and promotes recombination³⁵. On a more global scale, recent ChIP–chip analysis has demonstrated that THO components preferentially associate with active open reading frames (ORFs) genome-wide⁴⁷. In the absence of the THO complex, replisome movement is slower through these regions, and RNase H overexpression suppresses this effect.

THO complex

A conserved protein complex that includes the proteins Tho2, Hpr1, Mft1 and Thp2 in yeast. It interacts with the TREX complex and functions in mRNA metabolism and export.

Box 2 | **Programmed instability in the immune system**

Somatic hypermutation

Antigen-induced stimulation of B cells activates programmed mutagenesis of the heavy and light chain variable segments of immunoglobulin genes¹²⁴, which results in the production of high-affinity antibodies during an immune response. Transcription is crucial during somatic hypermutation (SHM) because it allows the single-strand-specific enzyme AID to access and deaminate cytosines on the non-transcribed strand. Uracils in the resulting U-G mispairs are excised by a uracil DNA glycosylase, leading to the accumulation of abasic sites. Mutations are introduced either during the gap-filling reaction that follows the excision of abasic sites or when replication bypasses the non-informative abasic sites. Highly mutagenic translesion synthesis DNA polymerases are probably used during gap-filling as well as during abasic site bypass. AID-dependent mutagenesis can also occur at other highly expressed, non-immunoglobulin genes in hypermutating B cells.

Class-switch recombination

Class-switch recombination (CSR) occurs between GC-rich switch regions that precede the constant segments of heavy-chain genes, which specify different classes of immunoglobulins (such as IgA and IgG). CSR requires AID, transcription and components of the non-homologous end-joining (NHEJ) pathway. Current models suggest that transcription-dependent, AID-initiated double-strand breaks form in two switch regions and that loss of the intervening DNA occurs when the broken ends are ligated by NHEJ. This irreversible process joins a new constant segment to the variable region of a heavy-chain gene. Interestingly, the breakpoints of NHEJ-derived translocations in B cells preferentially localize to transcribed regions of the genome¹²⁵.

A recent genome-wide screen in yeast has further broadened the involvement of RNA biogenesis in R-loop formation beyond the THO–TREX complex and further suggests that the associated DNA damage may sometimes occur outside the context of DNA replication⁴⁸. Finally, in the actively transcribed rDNA locus of yeast, both R-loops and extended regions of single-stranded DNA accumulate in the absence of Top1 and impede transcription^{49,50}. Importantly, loss of Top1 is associated with the accumulation of extrachromosomal rDNA circles formed via homologous recombination⁵¹, providing an additional connection between R-loops and TAR.

In higher eukaryotes, the relevance of R-loop formation to genome instability was first documented in chicken DT40 cells, in which depletion of the splicing factor serine/arginine-rich splicing factor 1 (SRSF1; also known as ASF1) resulted in the accumulation of R-loops and elevated genome rearrangements⁵². A genome-wide screen for functions that are relevant to genome maintenance in human cells has uncovered roles for diverse RNA-processing factors, and perturbations in RNA biogenesis were again associated with R-loop formation⁵³. Although the THO–TREX complex was not picked up in this analysis, directed depletion of individual components in human cells has demonstrated evolutionary conservation of its role in RNP biogenesis and genome stability⁵⁴.

A potential way of minimizing the genome-destabilizing effects of R-loops in eukaryotes is to separate transcription and replication temporally during the cell cycle. However, this is not an option for very large genes, within which complete transcription requires more than a single cell cycle. Indeed, a subset of common fragile sites of human chromosome breakage map to very long genes, and fragility has been associated with transcription and R-loop formation during S phase⁵⁵.

In the context of maintaining a stable genome, R-loops are clearly pathological structures. It should be noted, however, that there are at least two instances in which these structures are physiologically relevant. In bacteria, the RNA component of an R-loop is typically used as a primer to initiate DNA synthesis, and it could have a similar function in restarting replication forks in eukaryotes. In addition, R-loops are important in the vertebrate immune system, where they are proposed to have a role in facilitating class-switch recombination⁵⁶ (BOX 2).

Transcription and non-B-DNA structures

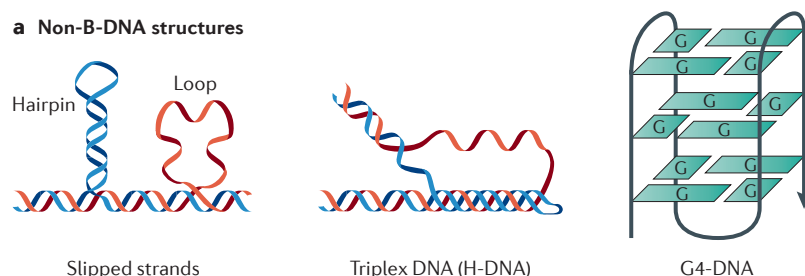
All DNA transactions (namely, replication, repair, recombination and transcription) require the transient separation of complementary strands, providing the opportunity for single-stranded DNA to assume non-canonical, non-B-DNA structures (FIG. 3a). During transcription, the single-stranded character of the negatively supercoiled region that accumulates behind RNAP can also facilitate the formation of non-B-DNA structures. Likewise, R-loops facilitate the accumulation of non-canonical DNA structures on the NTS, and non-B-DNA on the NTS promotes and stabilizes R-loops. In this section, we consider the destabilizing potential of two types of sequences that are known to form non-B-DNA structures: guanine-rich sequences and trinucleotide repeats.

Co-transcriptional G4-DNA and TAR. G-rich sequences can form stable, non-B structures known as G-quadruplex or G4-DNA, which is comprised of a stacked array of G quartets (FIG. 3a); these are planar structures in which four guanines are paired through Hoogsteen hydrogen bonds⁵⁷. An involvement of G4-DNA in genome instability is suggested by class-switch recombination in the vertebrate immune system — a process that requires transcription and that occurs between GC-rich ‘switch’ regions that are adjacent to the constant region gene segments of the immunoglobulin heavy-chain locus (BOX 2). A distinguishing feature of the switch regions is the asymmetric distribution of guanines and cytosines between the two DNA strands; in the physiological orientation, the G-rich strand is the NTS. When switch regions are highly expressed *in vitro* or in bacterial cells, bubbles containing G4-DNA opposite an R-loop (called a ‘G-loop’) have been observed by electron microscopy⁵⁸. Co-transcriptional G-loops have been shown to block transcription *in vitro*⁵⁹ and presumably would be a potent block to replication as well. Consistent with possible relevance for R-loops and/or G-loops in class-switch recombination, inversion of the switch region relative to the promoter converts the G-rich strand to the transcription template and eliminates most class-switch recombination⁶⁰.

We recently introduced a murine switch-region fragment into the yeast genome and examined the effects of transcription and of fragment orientation on stability using recombination as a readout⁶¹. Recapitulating observations in the immune system, recombination was enhanced by transcription and was further elevated when the G-rich strand was the NTS. Consistent with a causal role for co-transcriptional R-loop and/or G-loop formation, loss of either Top1 or RNase H activity exacerbated

the instability of switch-region sequences. Similarly, studies in *E. coli* have demonstrated an orientation-dependent effect of a switch-region sequence on TAR, which was accompanied by replication fork slowing and was suppressed by RNase H overexpression⁶².

a Non-B-DNA structures



b Expansion and contraction of CAG repeats

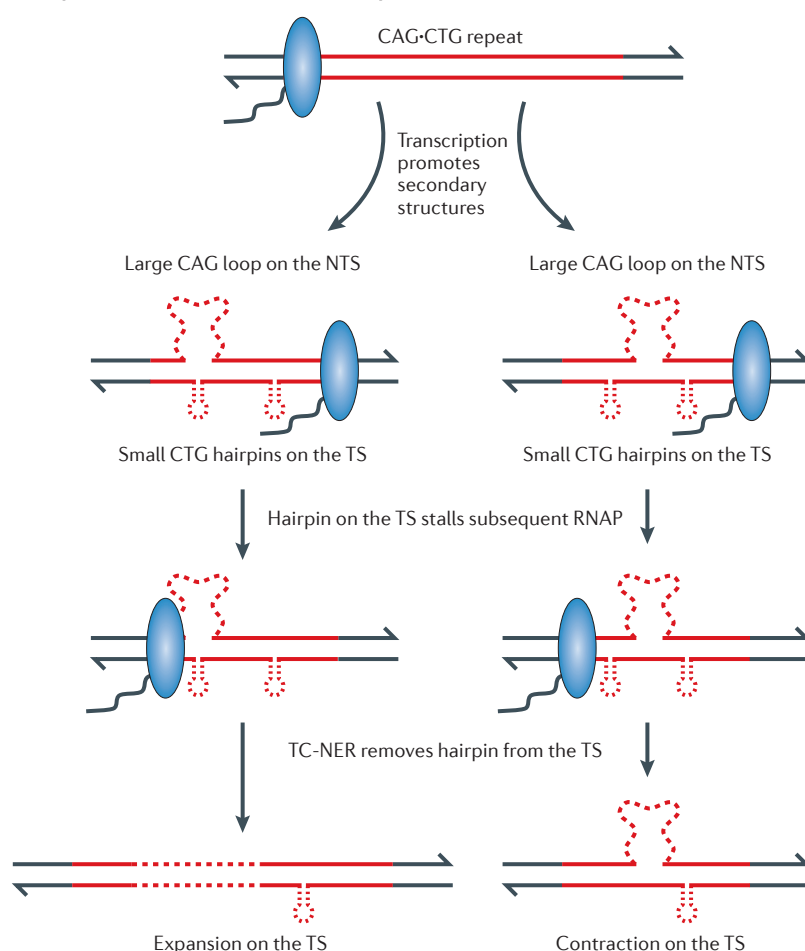


Figure 3 | Non-B-DNA structures and genome instability. **a** | Representative non-B-DNA structures are illustrated. **b** | Transcription through CAG·CTG repeats promotes the formation of slipped-strand structures, which subsequently stall RNA polymerase (RNAP) and lead to recruitment of the nucleotide excision repair (NER) machinery. Transcription-coupled NER removes the portion of the transcribed strand ('TS' in the figure) containing the RNAP-blocking hairpin; the resulting gap is filled in using the non-transcribed strand (NTS) as a template. Depending on the location of loops on the NTS relative to the removed hairpin, the repair event will either expand or contract the trinucleotide repeat.

Is G4-DNA relevant to transcription-associated genome instability beyond the specialized case of class-switch recombination? It seems likely, although only the destabilizing effect of G4-DNA in the context of DNA replication has been examined to date. For example, the GC-rich human *CEB1* (also known as *HERC5*) minisatellite, which efficiently forms G4-DNA *in vitro*, is highly unstable when it is introduced into the yeast genome. *CEB1* becomes more unstable following loss of the Pif1 helicase, which is capable of disrupting the G4-DNA formed by *CEB1* *in vitro*⁶³. In addition, ChIP-chip analysis has uncovered sequences that have G4-DNA-forming potential as preferential sites of yeast Pif1 association and as sites of replication slowing⁶⁴.

Instability of trinucleotide repeats. The expansion of trinucleotide repeats (TNRs) is responsible for at least 20 neurodegenerative and neuromuscular diseases, including fragile X syndrome (CGG·CCG), Huntington's disease (CAG·CTG) and Friedrich's ataxia (GAA·TTC)⁶⁵. Importantly, only those TNRs that are capable of forming non-B-DNA structures have been implicated in disease, leading to the belief that their formation is a crucial early step that initiates instability⁶⁶. Large germline expansions of TNRs have been most extensively studied and are thought to arise during replication; such expansions account for the initial disease appearance and the increase in severity in subsequent generations — a phenomenon that is referred to as 'anticipation'. TNR expansions that occur in somatic cells have attracted less attention but may be relevant to disease severity and progression, especially in non-dividing tissue⁶⁷. Expansions in non-dividing cells are, by definition, replication-independent, and recent work exploring transcription-associated TNR instability will briefly be summarized below.

Transcriptional effects on TNR instability have been characterized in human cell lines using CAG·CTG repeats, which can form stable slipped-hairpin structures⁶⁸ (FIG. 3a). Such structures on either the transcribed strand or the NTS are sufficient to stall RNAP *in vitro*⁶⁹. Key to the analysis of CAG·CTG instability in human cells has been the development of a transcriptionally regulated reporter that allows the direct selection of contraction events⁷⁰. Induction of transcription elevates contractions approximately 15-fold in this system, and this property has been used to identify proteins that enhance or suppress instability. When considered together, the data are consistent with the model shown in FIG. 3b, in which transcription promotes the formation of hairpins, which can stall the passage of subsequent RNAPs to trigger 'gratuitous' transcription-coupled nucleotide excision repair (TC-NER). Depending on the location of slip-outs relative to the stalled RNAP, the repair process can lead to either contractions or expansions of CAG·CTG repeats^{69,71}.

Only contractions can be selected in the mammalian experimental system noted above, but a *Drosophila melanogaster*-based system has been used to identify transcription-associated expansions of CAG·CTG repeats, and these likewise depend on TC-NER⁷². It seems likely that similar transcription-based instability

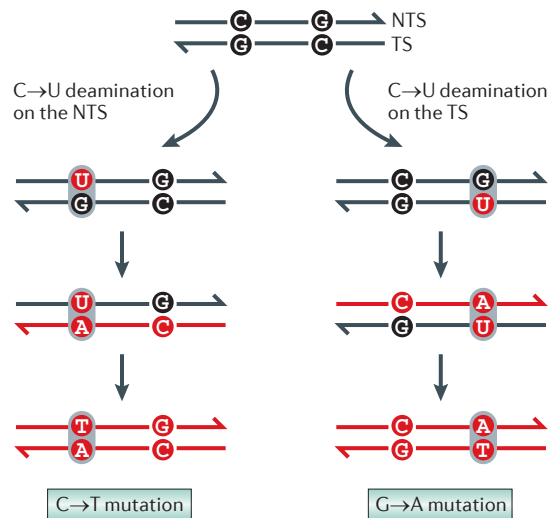


Figure 4 | Deducing the strand on which mutations arise. In the absence of uracil removal, deamination of C on the non-transcribed strand (NTS) leads to C→T mutations (note that by convention, DNA sequences are read from the NTS, which has the same sequence as the mRNA). By contrast, deamination of C on the transcribed strand ('TS' in the figure) results in G→A mutations.

mechanisms will also apply to other classes of TNRs. GAA·TTC repeats, for example, have the propensity to form three-stranded H-DNA or triplex DNA⁷³ (FIG. 3a); they promote R-loop formation in bacterial cells⁷⁴, and they are destabilized by transcription in mammalian cells^{75,76}. To date, there have been no reports of transcriptional effects on CGG·CCG stability.

Analyses of expressed sequences in mammalian cells indicate that both strands of DNA are often transcribed⁷⁷. Therefore, recent work has examined the effect of simultaneously transcribing through CAG·CTG repeats in both directions⁷⁸. Importantly, convergent transcription led to greater instability than would be predicted by summing the individual effects of forward and reverse transcription^{79,80} and was an efficient trigger of apoptosis⁷⁹.

Transcription-associated DNA damage and TAM

The disabling of DNA damage repair or bypass pathways in yeast elevates spontaneous TAM in defined reversion assays, implicating transcription-associated damage as a causative factor^{3,81}. Striking synergistic effects of transcription and exogenous mutagens on TAR have also been reported, which is again consistent with an enhanced accumulation of damage in highly transcribed DNA⁸². Unexpectedly, recent work has also found that Top1 activity, which reduces R-loop-associated TAR, can be an important source of TAM^{83,84}.

Chemical modification of DNA biases TAM to the NTS. In reversion assays, all types of base substitutions appear to be stimulated by transcription in yeast⁸⁵ and bacterial cells^{85–87}; additionally, a large variety of transcription-associated insertions or deletions have been detected

in yeast^{7,81}. The strand-of-origin of a given mutation is impossible to deduce in wild-type cells, but it can be assigned with confidence in some repair-defective backgrounds. In the absence of uracil DNA glycosylase, for example, CG→TA transitions can be assumed to result from cytosine deamination to uracil rather than from damage to guanine on the complementary strand (FIG. 4). In *E. coli*, such mutations are strongly biased towards cytosines that are located on the NTS when transcription is highly activated^{188,89}, as are G→T mutations that are associated with oxidation or loss of guanine⁹⁰. Because chemical modifications occur much more often in single-stranded DNA than they do in duplex DNA⁹¹, it has been argued that the enhanced single-stranded character of the NTS is an important contributor to TAM.

An NTS-related bias for spontaneous TAM has not been examined in yeast, but it has been demonstrated that enzymatic deamination of cytosine to uracil primarily targets cytosines located on the NTS of active genes in THO mutants⁹². Such deamination is required for the somatic hypermutation of immunoglobulin genes; this is a specialized process in which the initiating event primarily accesses the NTS through a transcription-dependent process (BOX 2). A much more general, strand-specific effect of transcription has been inferred through whole-genome sequencing. A comparative analysis of nine mammalian genomes, for example, reported a bias for A→G relative to T→C mutations on the NTS⁹³, and the degree of asymmetry correlates with the gene expression level in the germline⁹⁴. More recently, an examination of somatic 'passenger' mutations in rapidly evolving tumour cells has suggested that a similar asymmetry is generated during somatic divisions⁹⁵.

The strand asymmetries that are associated with TAM could reflect the transient exposure of small regions of the NTS within the transcription bubble that moves with RNAP and/or the formation of more extensive regions behind the transcription machinery. Although the negative supercoils formed in the wake of RNAP would be expected to impart single-stranded character to both strands, the associated formation of R-loops would confine the exposed DNA to the NTS. A more refined model has been proposed in which secondary structures formed on the NTS only leave specific bases exposed in an unpaired and vulnerable state⁹⁶. An algorithm has been developed to predict the most likely sites of potential damage, and its predictive value has been demonstrated using a bacterial reporter⁹⁷. Intriguingly, this analysis has been extended to the human *TP53* gene (which encodes p53), in which the predicted frequency of base exposure positively correlates with the frequency of mutation at the 12 most mutable *TP53* sites in human tumours⁹⁸.

An alternative but not necessarily mutually exclusive model for the biased accumulation of mutations on the NTS is that TC-NER preferentially removes lesions from the transcribed strand⁹⁹. A TC-NER-dependent modulation in mutation spectra, for example, has been reported for *E. coli* treated with alkylating agents¹⁰⁰ or exposed to ultraviolet irradiation¹⁰¹. In yeast, we recently demonstrated that TC-NER likewise alters

Transcription-coupled nucleotide excision repair (TC-NER). TC-NER is a subpathway of the NER pathway that is initiated specifically in response to an RNA polymerase arrested by damage on the DNA template. The net effect is more efficient NER-directed repair of lesions on the transcribed than on the non-transcribed strand of active genes.

Reversion assays

Assays that start with a mutant allele, typically containing a change in a single base pair or the insertion or deletion of a single base pair, and that select for restoration of gene function. The change that restores gene function is usually limited to the position of the original mutation.

spontaneous mutation patterns in a highly transcribed gene⁸⁵. Finally, TC-NER has been invoked to explain a mutagen-induced NTS bias for mutations in the *TP53* tumour suppressor gene in human cells¹⁰². Although it is generally accepted that transcription biases mutations to the NTS, a reversal in this pattern has been reported following the treatment of mouse embryonic stem cells (ESCs) with ultraviolet light^{103,104}.

Top1 activity as a source of TAM. In repair-competent yeast cells, the class of forward mutations that are increased most by transcription is comprised of short deletions (2–5 bp). These signature deletions are uniquely dependent on the activity of Top1 and presumably reflect its recruitment to relieve the supercoiling associated with transcriptionally active DNA^{83,84}. Recent analyses suggest that Top1-dependent mutations have two distinct causes (BOX 1). The first reflects an irreversible trapping of Top1 on DNA during its normal cleavage-ligation cycle, whereas the second reflects Top1-mediated cleavage at a ribonucleotide monophosphate (rNMP) that has been misincorporated into DNA¹⁰⁵. Whether both types of Top1 cleavage product generate the observed deletions through a common intermediate or via distinct mechanisms is not known.

Transcription and change in DNA composition

We recently reported that transcription in yeast alters the nucleotide composition of the underlying template in an unexpected way: specifically increasing the direct incorporation of deoxyuridine monophosphate (dUMP) in place of deoxythymidine monophosphate (dTTP)¹⁰⁶. This was discovered through genetic analyses of a distinctive TAM signature that is associated with reduced efficiency of the base excision repair pathway, which is specialized to repair abasic sites. Importantly, the TAM signature required an ability to excise uracil from the DNA backbone, indicating that uracil levels increase in highly transcribed DNA and that its removal is responsible for most abasic sites. Although the excess uracil could be derived from cytosine deamination, TAM dramatically decreased following overexpression of the yeast deoxyuridine triphosphatase (dUTPase) — an enzyme that specifically hydrolyses dUTP. Because this lowers the dUTP/deoxythymidine triphosphate (dTTP) ratio in the nucleotide pool and thereby disfavours dUTP incorporation into DNA, the results imply that the elevated uracil is derived from direct dUTP incorporation associated with transcription.

In higher eukaryotes, the removal of uracils from the DNA backbone is a crucial and early step in the molecular events that are associated with somatic hypermutation and class-switch recombination within immunoglobulin genes (BOX 2). Although directed cytosine deamination by the enzyme AID is the major source of uracil during these processes, direct incorporation of dUTP by DNA polymerase has not yet been ruled out as a minor contributor¹⁰⁷. It was recently reported, for example, that uracil can be detected at positions of thymines in the variable region of mouse immunoglobulin genes¹⁰⁸. However, another study concluded that uracil in

hypermutating immunoglobulin genes was only found at positions of cytosines¹⁰⁹. Given the results in yeast and the generally universal effects of transcription on genome instability, it is expected that direct incorporation of dUTP into highly transcribed DNA will extend to other systems.

Precisely how transcription affects the nucleotide pool remains to be answered. One possibility is that a locally higher, transcription-associated dUTP concentration forces more frequent uracil incorporation by DNA polymerases. Spatially, a subnuclear localization of highly transcribed genes might be involved. Temporally, unscheduled DNA synthesis occurring outside S phase, such as that involved in the repair of transcription-associated damage, might be responsible for the elevated dUTP incorporation. It is known, for example, that expression of the yeast dUTPase gene is upregulated during S phase¹¹⁰, thereby lowering dUTP levels within the dNTP pool and reducing its incorporation into DNA during genome duplication.

Conclusions and future directions

In this Review, we have summarized the current knowledge of the diverse ways in which transcription locally stimulates recombination and/or mutagenesis: through collisions with the replication machinery, formation of co-transcriptional R-loops, facilitation of non-B-DNA structure formation, engagement of Top1 activity, promotion of DNA damage and alterations in DNA base composition. In future studies, it will be important to determine how conserved a given mechanism is and to determine its contribution relative to other transcription-related mechanisms that affect genome stability. Even if a given mechanism is a fairly rare contributor to TAM and/or TAR in 'normal' cells, its importance may shift under some conditions. This may be particularly relevant during the evolution of tumours, where there may already be underlying deficiencies in the DNA repair and checkpoint processes. In addition, both the levels of endogenous mutagens and global transcription profiles are expected to fluctuate in response to environmental conditions. Environmental influences on transcription have the potential to alter the evolutionary landscape by targeting changes to genes where they are most likely to be beneficial in terms of promoting growth. In bacterial cells, transcription thus provides a mechanism for adaptive or stress-induced mutagenesis^{8,111}; in higher eukaryotes, a similar process could contribute to tumour evolution. An important take-home lesson from the results obtained to date is that transcription probably alters the genetic landscape of all organisms on an evolutionary timescale.

In closing, we would like to mention other connections between transcription and genome stability that were not specifically dealt with in this Review. In addition to coupling of nucleotide excision repair pathway to transcription (TC-NER), proteins that are involved in both single- and double-strand break repair have been isolated as part of a complex with human RNAP II (REF. 112). Are these proteins possibly there to deal specifically with damage caused by transcription, or is

Forward mutations

Forward mutation assays select for loss of a gene function and can detect any change in the DNA sequence that inactivates the encoded product, which is usually a protein.

transcription (in the form of damage-blocked RNAP complexes) used as a sensor to target repair where it is most urgently needed for proper gene expression? Alternatively, does transcription provide an efficient way of detecting and dealing with damage that can potentially impede DNA polymerases before the damage actually causes a replication-associated problem?

A potential twist in the TAR story comes from studies in budding yeast demonstrating that the DNA synthesis that is associated with homologous recombination is much more error-prone than that associated with genome duplication^{113–115}. An intriguing possibility is that some TAM might simply be a by-product of DNA synthesis that is associated with TAR. Although most of the stimulatory effects of transcription on homologous recombination are likely to come through the generation of recombination-initiating lesions¹¹⁶, this is not the entire story. Transcription additionally seems to enhance access to the repair template in both budding yeast and human cells^{117,118}. In addition to the process of transcription, the primary transcript itself may be relevant to recombination processes. Studies

in yeast have demonstrated, for example, that spliced transcripts can be reverse-transcribed by endogenous enzymes into cDNAs, which can then be used to correct chromosomal mutations¹¹⁹. Such a mechanism has been invoked to explain why most yeast genes are devoid of introns¹²⁰; in metazoans, random rather than homology-based insertion of cDNAs has for a long time been assumed to be the source of intron-less and promoter-less pseudogenes. Finally, studies have shown that small RNAs transformed into yeast cells can directly be used to template the repair of chromosomal double-strand breaks¹²¹.

With respect to TAM, only the tip of the iceberg has been touched in terms of what types of damage accumulate in highly transcribed DNA. Importantly, TAM can provide a replication-independent source of mutations that may be particularly relevant in post-mitotic cells or under defined starvation conditions. We expect and look forward to the discovery of additional mechanisms of transcription-associated mutagenesis and recombination as more systems are developed to study these fascinating processes.

1. Savic, D. J. & Kanazir, D. T. The effect of a histidine operator-constitutive mutation on UV-induced mutability within the histidine operon of *Salmonella typhimurium*. *Mol. Gen. Genet.* **118**, 45–50 (1972).
2. Herman, R. K. & Dworkin, N. B. Effect of gene induction on the rate of mutagenesis by ICR-191 in *Escherichia coli*. *J. Bacteriol.* **106**, 543–550 (1971).
3. Datta, A. & Jinks-Robertson, S. Association of increased spontaneous mutation rates with high levels of transcription in yeast. *Science* **268**, 1616–1619 (1995).
- This work, using budding yeast as a model system, was the first demonstration of transcription-associated mutagenesis in eukaryotic cells.**
4. Saxowsky, T. T. & Doetsch, P. W. RNA polymerase encounters with DNA damage: transcription-coupled repair or transcriptional mutagenesis? *Chem. Rev.* **106**, 474–488 (2006).
5. Keil, R. L. & Roeder, G. S. Cis-acting, recombination-stimulating activity in a fragment of the ribosomal DNA of *S. cerevisiae*. *Cell* **39**, 377–386 (1984).
6. Voekel-Meiman, K., Keil, R. L. & Roeder, G. S. Recombination-stimulating sequences in yeast ribosomal DNA correspond to sequences regulating transcription by RNA polymerase I. *Cell* **48**, 1071–1079 (1987).
- This paper identifies the recombination-stimulating sequence *HOT1* as the rDNA promoter in budding yeast, thereby providing the first link between transcription and mitotic recombination.**
7. Kim, N., Abdulovic, A. L., Gealy, R., Lippert, M. J. & Jinks-Robertson, S. Transcription-associated mutagenesis in yeast is directly proportional to the level of gene expression and influenced by the direction of DNA replication. *DNA Repair* **6**, 1285–1296 (2007).
8. Reimers, J. M., Schmidt, K. H., Longacre, A., Reschke, D. K. & Wright, B. E. Increased transcription rates correlate with increased reversion rates in *leuB* and *argH* *Escherichia coli* auxotrophs. *Microbiology* **150**, 1457–1466 (2004).
9. Nickoloff, J. A. Transcription enhances intrachromosomal homologous recombination in mammalian cells. *Mol. Cell. Biol.* **12**, 5311–5318 (1992).
10. Postow, L. *et al.* Positive torsional strain causes the formation of a four-way junction at replication forks. *J. Biol. Chem.* **276**, 2790–2796 (2001).
11. Rudolph, C. J., Dhillon, P., Moore, T. & Lloyd, R. G. Avoiding and resolving conflicts between DNA replication and transcription. *DNA Repair* **6**, 981–993 (2007).
12. Pomerantz, R. T. & O'Donnell, M. What happens when replication and transcription complexes collide? *Cell Cycle* **9**, 2537–2543 (2010).
13. Gottipati, P. & Helleday, T. Transcription-associated recombination in eukaryotes: link between transcription, replication and recombination. *Mutagenesis* **24**, 203–210 (2009).
14. Ellwood, M. & Nomura, M. Chromosomal locations of the genes for rRNA in *Escherichia coli* K-12. *J. Bacteriol.* **149**, 458–468 (1982).
15. Guy, L. & Roten, C. A. Genomic analyses of the organization of circular chromosomes: a universal pressure determines the direction of ribosomal RNA genes transcription relative to chromosome replication. *Gene* **340**, 45–52 (2004).
16. Wang, J. D., Berkmen, M. B. & Grossman, A. D. Genome-wide coorientation of replication and transcription reduces adverse effects on replication in *Bacillus subtilis*. *Proc. Natl Acad. Sci. USA* **104**, 5608–5613 (2007).
17. Srivatsan, A., Tehranchi, A., MacAlpine, D. M. & Wang, J. D. Co-orientation of replication and transcription preserves genome integrity. *PLoS Genet.* **6**, e1000810 (2010).
18. Brewer, B. J. & Fangman, W. L. A replication fork barrier at the 3' end of yeast ribosomal RNA genes. *Cell* **55**, 637–643 (1988).
19. Huvel, M. *et al.* Human gene organization driven by the coordination of replication and transcription. *Genome Res.* **17**, 1278–1285 (2007).
20. Mirkin, E. V. & Mirkin, S. M. Mechanisms of transcription-replication collisions in bacteria. *Mol. Cell. Biol.* **25**, 888–895 (2005).
21. Deshpande, A. M. & Newlon, C. S. DNA replication fork pause sites dependent on transcription. *Science* **272**, 1030–1033 (1996).
22. Prado, F. & Aguilera, A. Impairment of replication fork progression mediates RNA polII transcription-associated recombination. *EMBO J.* **24**, 1267–1276 (2005).
23. Takeuchi, Y., Horiuchi, T. & Kobayashi, T. Transcription-dependent recombination and the role of fork collision in yeast rDNA. *Genes Dev.* **17**, 1497–1506 (2003).
24. Azvolinsky, A., Giresi, P. G., Lieb, J. D. & Zakian, V. A. Highly transcribed RNA polymerase II genes are impediments to replication fork progression in *Saccharomyces cerevisiae*. *Mol. Cell* **34**, 722–734 (2009).
- In this paper, the ChIP-chip technique was used to identify sites of DNA Pol II occupancy, which are presumed to reflect replication pausing, throughout the budding yeast genome. Highly transcribed ORFs were enriched among these sites, indicating that transcription can impede replication.**
25. Cox, M. M. *et al.* The importance of repairing stalled replication forks. *Nature* **404**, 37–41 (2000).
26. Gottipati, P., Cassel, T. N., Savolainen, L. & Helleday, T. Transcription-associated recombination is dependent on replication in mammalian cells. *Mol. Cell. Biol.* **28**, 154–164 (2008).
27. de la Loza, M. C., Wellinger, R. E. & Aguilera, A. Stimulation of direct-repeat recombination by RNA polymerase III transcription. *DNA Repair* **8**, 620–626 (2009).
28. Sikdar, N., Banerjee, S., Zhang, H., Smith, S. & Myung, K. Spt2p defines a new transcription-dependent gross chromosomal rearrangement pathway. *PLoS Genet.* **4**, e1000290 (2008).
29. Vilette, D., Ehrlich, S. D. & Michel, B. Transcription-induced deletions in *Escherichia coli* plasmids. *Mol. Microbiol.* **17**, 493–504 (1995).
30. Dutta, D., Shatalin, K., Epshtein, V., Gottesman, M. E. & Nudler, E. Linking RNA polymerase backtracking to genome instability in *E. coli*. *Cell* **146**, 533–543 (2011).
31. Blobel, G. Gene gating: a hypothesis. *Proc. Natl Acad. Sci. USA* **82**, 8527–8529 (1985).
32. Bermejo, R. *et al.* The replication checkpoint protects fork stability by releasing transcribed genes from nuclear pores. *Cell* **146**, 233–246 (2011).
33. Waters, L. S. *et al.* Eukaryotic translation polymerases and their roles and regulation in DNA damage tolerance. *Microbiol. Mol. Biol. Rev.* **73**, 134–154 (2009).
34. Aguilera, A. The connection between transcription and genomic instability. *EMBO J.* **21**, 195–201 (2002).
35. Li, X. & Manley, J. L. Cotranscriptional processes and their influence on genome stability. *Genes Dev.* **20**, 1838–1847 (2006).
36. Gowrishankar, J. & Harinarayanan, R. Why is transcription coupled to translation in bacteria? *Mol. Microbiol.* **54**, 598–603 (2004).
37. Wu, H. Y., Shyy, S. H., Wang, J. C. & Liu, L. F. Transcription generates positively and negatively supercoiled domains in the template. *Cell* **53**, 433–440 (1988).
- This paper proposed the 'twin domain' model of transcription-associated DNA helical stress.**
38. Tuduri, S. *et al.* Topoisomerase I suppresses genomic instability by preventing interference between replication and transcription. *Nature Cell Biol.* **11**, 1315–1324 (2009).
39. Cerritelli, S. M. & Crouch, R. J. Ribonuclease H: the enzymes in eukaryotes. *FEBS J.* **276**, 1494–1505 (2009).
40. Baaklini, I., Hraiky, C., Rallu, F., Tse-Dinh, Y. C. & Drolet, M. RNase H1 overproduction is required for efficient full-length RNA synthesis in the absence of topoisomerase I in *Escherichia coli*. *Mol. Microbiol.* **54**, 198–211 (2004).

41. Mischo, H. E. *et al.* Yeast Sen1 helicase protects the genome from transcription-associated instability. *Mol. Cell* **41**, 21–32 (2011).
42. Rudolph, C. J., Upton, A. L., Briggs, G. S. & Lloyd, R. G. Is RecG a general guardian of the bacterial genome? *DNA Repair* **9**, 210–223 (2010).
43. Skourtis-Stathaki, K., Proudfoot, N. J. & Gromak, N. Human senataxin resolves RNA/DNA hybrids formed at transcriptional pause sites to promote Xrn2-dependent termination. *Mol. Cell* **42**, 794–805 (2011).
44. Chavez, S. & Aguilera, A. The yeast *HPR1* gene has a functional role in transcriptional elongation that uncovers a novel source of genome instability. *Genes Dev.* **11**, 3459–3470 (1997).
45. Huertas, P. & Aguilera, A. Cotranscriptionally formed DNA:RNA hybrids mediate transcription elongation impairment and transcription-associated recombination. *Mol. Cell* **12**, 711–721 (2003).
This paper highlights the role of co-transcriptionally formed RNA–DNA hybrids (R-loops) in TAR by showing that degradation of the nascent mRNA by RNase H1 or a self-cleaving hammerhead ribozyme can suppress hyper-recombination in budding yeast *hpr1* mutants.
46. Rondon, A. G., Jimeno, S. & Aguilera, A. The interface between transcription and mRNP export: from THO to THSC/TREX-2. *Biochim. Biophys. Acta* **1799**, 533–538 (2010).
47. Gomez-Gonzalez, B. *et al.* Genome-wide function of THO/TREX in active genes prevents R-loop-dependent replication obstacles. *EMBO J.* **30**, 3106–3119 (2011).
48. Wahba, L., Amon, J. D., Koshland, D. & Vuica-Ross, M. RNase H and multiple RNA biogenesis factors cooperate to prevent RNA:DNA hybrids from generating genome instability. *Mol. Cell* **44**, 978–988 (2011).
49. French, S. L. *et al.* Distinguishing the roles of Topoisomerases I and II in relief of transcription-induced torsional stress in yeast rRNA genes. *Mol. Cell. Biol.* **31**, 482–494 (2011).
50. El Hage, A., French, S. L., Beyer, A. L. & Tollervy, D. Loss of Topoisomerase I leads to R-loop-mediated transcriptional blocks during ribosomal RNA synthesis. *Genes Dev.* **24**, 1546–1558 (2010).
51. Christman, M. F., Dietrich, F. S. & Fink, G. R. Mitotic recombination in the rDNA of *S. cerevisiae* is suppressed by the combined action of DNA topoisomerases I and II. *Cell* **55**, 413–425 (1988).
52. Li, X. & Manley, J. L. Inactivation of the SR protein splicing factor ASF/SF2 results in genomic instability. *Cell* **122**, 365–378 (2005).
This paper demonstrated the role of mRNA splicing complex assembly in suppressing TAR in chicken DT40 and mammalian cells.
53. Paulsen, R. D. *et al.* A genome-wide siRNA screen reveals diverse cellular processes and pathways that mediate genome stability. *Mol. Cell* **35**, 228–239 (2009).
54. Dominguez-Sanchez, M. S., Barroso, S., Gomez-Gonzalez, B., Luna, R. & Aguilera, A. Genome instability and transcription elongation impairment in human cells depleted of THO/TREX. *PLoS Genet.* **7**, e1002386 (2011).
55. Helmrich, A., Ballarino, M. & Tora, L. Collisions between replication and transcription complexes cause common fragile site instability at the longest human genes. *Mol. Cell* **44**, 966–977 (2011).
56. Yu, K., Chedin, F., Hsieh, C. L., Wilson, T. E. & Lieber, M. R. R-loops at immunoglobulin class switch regions in the chromosomes of stimulated B cells. *Nature Immunol.* **4**, 442–451 (2003).
57. Maizels, N. Dynamic roles for G4 DNA in the biology of eukaryotic cells. *Nature Struct. Mol. Biol.* **13**, 1055–1059 (2006).
58. Duquette, M. L., Handa, P., Vincent, J. A., Taylor, A. F. & Maizels, N. Intracellular transcription of G-rich DNAs induces formation of G-loops, novel structures containing G4 DNA. *Genes Dev.* **18**, 1618–1629 (2004).
This work demonstrated that a highly transcribed immunoglobulin switch region sequence forms G-loop structures visible by electron microscope both *in vitro* and in cells in *E. coli*.
59. Belotserkovskii, B. P. *et al.* Mechanisms and implications of transcription blockage by guanine-rich DNA sequences. *Proc. Natl Acad. Sci. USA* **107**, 12816–12821 (2010).
60. Shinkura, R. *et al.* The influence of transcriptional orientation on endogenous switch region function. *Nature Immunol.* **4**, 435–441 (2003).
61. Kim, N. & Jinks-Robertson, S. Guanine repeat-containing sequences confer transcription-dependent instability in an orientation-specific manner in yeast. *DNA Repair* **10**, 953–960 (2011).
62. Gan, W. *et al.* R-loop-mediated genomic instability is caused by impairment of replication fork progression. *Genes Dev.* **25**, 2041–2056 (2011).
63. Lopes, J. *et al.* G-quadruplex-induced instability during leading-strand replication. *EMBO J.* **30**, 4033–4046 (2011).
64. Paeschke, K., Capra, J. A. & Zakian, V. A. DNA replication through G-quadruplex motifs is promoted by the *Saccharomyces cerevisiae* Pif1 DNA helicase. *Cell* **145**, 678–691 (2011).
65. Mirkin, S. M. Expandable DNA repeats and human disease. *Nature* **447**, 932–940 (2007).
66. McMurray, C. T. DNA secondary structure: a common and causative factor for expansion in human disease. *Proc. Natl Acad. Sci. USA* **96**, 1823–1825 (1999).
67. Swami, M. *et al.* Somatic expansion of the Huntington's disease CAG repeat in the brain is associated with an earlier age of disease onset. *Hum. Mol. Genet.* **18**, 3039–3047 (2009).
68. Petruska, J., Arnheim, N. & Goodman, M. F. Stability of intrastrand hairpin structures formed by the CAG/CTG class of DNA triplet repeats associated with neurological diseases. *Nucleic Acids Res.* **24**, 1992–1998 (1996).
69. Salinas-Rios, V., Belotserkovskii, B. P. & Hanawalt, P. C. DNA slip-outs cause RNA polymerase II arrest *in vitro*: potential implications for genetic instability. *Nucleic Acids Res.* **39**, 7444–7454 (2011).
70. Lin, Y., Dion, V. & Wilson, J. H. Transcription promotes contraction of CAG repeat tracts in human cells. *Nature Struct. Mol. Biol.* **13**, 179–180 (2006).
This work showed that contraction of CAG triplet repeats in human cells requires transcription and components of defined DNA repair pathways.
71. Lin, Y. & Wilson, J. H. Transcription-induced CAG repeat contraction in human cells is mediated in part by transcription-coupled nucleotide excision repair. *Mol. Cell. Biol.* **27**, 6209–6217 (2007).
72. Jung, J. & Bonini, N. CREB-binding protein modulates repeat instability in a *Drosophila* model for polyQ disease. *Science* **315**, 1857–1859 (2007).
Using a *D. melanogaster* model system, this work demonstrated that CAG repeat expansion requires transcription and is mediated by TC-NER.
73. Wells, R. D., Dere, R., Hebert, M. L., Napierala, M. & Son, L. S. Advances in mechanisms of genetic instability related to hereditary neurological diseases. *Nucleic Acids Res.* **33**, 3785–3798 (2005).
74. Grabczyk, E., Mancuso, M. & Sammarco, M. C. A persistent RNA-DNA hybrid formed by transcription of the Friedreich ataxia triplet repeat in live bacteria, and by T7 RNAP *in vitro*. *Nucleic Acids Res.* **35**, 5351–5359 (2007).
75. Ditch, S., Sammarco, M. C., Banerjee, A. & Grabczyk, E. Progressive GAA-TTC repeat expansion in human cell lines. *PLoS Genet.* **5**, e1000704 (2009).
76. Rindler, P. M. & Bidichandani, S. I. Role of transcript and interplay between transcription and replication in triplet-repeat instability in mammalian cells. *Nucleic Acids Res.* **39**, 526–535 (2011).
77. Yelin, R. *et al.* Widespread occurrence of antisense transcription in the human genome. *Nature Biotech.* **21**, 379–386 (2003).
78. Lin, Y. & Wilson, J. H. Transcription-induced DNA toxicity at trinucleotide repeats: double bubble is trouble. *Cell Cycle* **10**, 611–618 (2011).
79. Lin, Y., Leng, M., Wan, M. & Wilson, J. H. Convergent transcription through a long CAG tract destabilizes repeats and induces apoptosis. *Mol. Cell. Biol.* **30**, 4435–4451 (2010).
80. Nakamori, M., Pearson, C. E. & Thornton, C. A. Bidirectional transcription stimulates expansion and contraction of expanded (CTG)(CAG) repeats. *Hum. Mol. Genet.* **20**, 580–588 (2011).
81. Morey, N. J., Greene, C. N. & Jinks-Robertson, S. Genetic analysis of transcription-associated mutation in *Saccharomyces cerevisiae*. *Genetics* **154**, 109–120 (2000).
82. Garcia-Rubio, M., Huertas, P., Gonzalez-Barrera, S. & Aguilera, A. Recombinogenic effects of DNA-damaging agents are synergistically increased by transcription in *Saccharomyces cerevisiae*. New insights into transcription-associated recombination. *Genetics* **165**, 457–466 (2003).
83. Lippert, M. J. *et al.* Role for topoisomerase 1 in transcription-associated mutagenesis in yeast. *Proc. Natl Acad. Sci. USA* **108**, 698–703 (2011).
84. Takahashi, T., Burguiere-Slezak, G., Van der Kemp, P. A. & Boiteux, S. Topoisomerase 1 provokes the formation of short deletions in repeated sequences upon high transcription in *Saccharomyces cerevisiae*. *Proc. Natl Acad. Sci. USA* **108**, 692–697 (2011).
The above two references describe transcription-dependent short deletions at tandem repeats and demonstrate that these events require Top1 activity.
85. Kim, N. & Jinks-Robertson, S. Abasic sites in the transcribed strand of yeast DNA are removed by transcription-coupled nucleotide excision repair. *Mol. Cell. Biol.* **30**, 3206–3215 (2010).
This paper showed that uracil-derived AP sites and the ensuing mutagenesis are highly stimulated by transcription in budding yeast. The uracil reflects direct incorporation of dUTP in place of dTTP rather than cytosine deamination.
86. Hudson, R. E., Berghthorsson, U. & Ochman, H. Transcription increases multiple spontaneous point mutations in *Salmonella enterica*. *Nucleic Acids Res.* **31**, 4517–4522 (2003).
87. Klapacz, J. & Bhagwat, A. S. Transcription-dependent increase in multiple classes of base substitution mutations in *Escherichia coli*. *J. Bacteriol.* **184**, 6866–6872 (2002).
88. Beletskii, A. & Bhagwat, A. S. Transcription-induced mutations: increase in C to T mutations in the nontranscribed strand during transcription in *Escherichia coli*. *Proc. Natl Acad. Sci. USA* **93**, 13919–13924 (1996).
This work demonstrated that in *E. coli*, C to T mutations resulting from spontaneous deamination of cytosines occur more frequently on the NTS of an active gene.
89. Beletskii, A. & Bhagwat, A. S. Correlation between transcription and C to T mutations in the non-transcribed DNA strand. *Biol. Chem.* **379**, 549–551 (1998).
90. Klapacz, J. & Bhagwat, A. S. Transcription promotes guanine to thymine mutations in the non-transcribed strand of an *Escherichia coli* gene. *DNA Repair* **4**, 806–813 (2005).
91. Frederico, L. A., Kunkel, T. A. & Shaw, B. R. A sensitive genetic assay for the detection of cytosine deamination: determination of rate constants and the activation energy. *Biochemistry* **29**, 2532–2537 (1990).
92. Gomez-Gonzalez, B. & Aguilera, A. Activation-induced cytidine deaminase action is strongly stimulated by mutations of the THO complex. *Proc. Natl Acad. Sci. USA* **104**, 8409–8414 (2007).
93. Green, P., Ewing, B., Miller, W., Thomas, P. J. & Green, E. D. Transcription-associated mutational asymmetry in mammalian evolution. *Nature Genet.* **33**, 514–517 (2003).
94. Majewski, J. Dependence of mutational asymmetry on gene-expression levels in the human genome. *Am. J. Hum. Genet.* **73**, 688–692 (2003).
The above two papers together demonstrated a strand-specific mutation bias in mammalian genes that correlates with expression level.
95. Rubin, A. F. & Green, P. Mutation patterns in cancer genomes. *Proc. Natl Acad. Sci. USA* **106**, 21766–21770 (2009).
96. Wright, B. E., Reschke, D. K., Schmidt, K. H., Reimers, J. M. & Knight, W. Predicting mutation frequencies in stem-loop structures of derepressed genes: implications for evolution. *Mol. Microbiol.* **48**, 429–441 (2003).
97. Schmidt, K. H., Reimers, J. M. & Wright, B. E. The effect of promoter strength, supercoiling and secondary structure on mutation rates in *Escherichia coli*. *Mol. Microbiol.* **60**, 1251–1261 (2006).
98. Wright, B. E. *et al.* The roles of transcription and genotoxins underlying p53 mutagenesis *in vivo*. *Carcinogenesis* **32**, 1559–1567 (2011).
99. Hanawalt, P. C. & Spivak, G. Transcription-coupled DNA repair: two decades of progress and surprises. *Nature Rev. Mol. Cell Biol.* **9**, 958–970 (2008).
100. Iannone, R. *et al.* Mutation spectra analysis suggests that N-(2-chloroethyl)-N'-nitrosourea-induced lesions are subject to transcription-coupled repair in *Escherichia coli*. *Mol. Carcinog.* **19**, 39–45 (1997).
101. Li, B. H., Ebbert, A. & Bockrath, R. Transcription-modulated repair in *Escherichia coli* evident with UV-induced mutation spectra in *supF*. *J. Mol. Biol.* **294**, 35–48 (1999).

102. Moriya, M. *et al.* TP53 Mutational signature for aristolochic acid: an environmental carcinogen. *Int. J. Cancer* **129**, 1532–1536 (2011).
103. Hendriks, G. *et al.* Gene transcription increases DNA damage-induced mutagenesis in mammalian stem cells. *DNA Repair* **7**, 1330–1339 (2008).
104. Hendriks, G. *et al.* Transcription-dependent cytosine deamination is a novel mechanism in ultraviolet light-induced mutagenesis. *Curr. Biol.* **20**, 170–175 (2010).
105. Kim, N. *et al.* Mutagenic processing of ribonucleotides in DNA by yeast topoisomerase I. *Science* **332**, 1561–1564 (2011).
106. Kim, N. & Jinks-Robertson, S. dUTP incorporation into genomic DNA is linked to transcription in yeast. *Nature* **459**, 1150–1153 (2009).
107. Neuberger, M. S. *et al.* Somatic hypermutation at A.T pairs: polymerase error versus dUTP incorporation. *Nature Rev. Immunol.* **5**, 171–178 (2005).
108. Roche, B., Claes, A. & Rougeon, F. Deoxyuridine triphosphate incorporation during somatic hypermutation of mouse V α Ox genes after immunization with phenylloxazalone. *J. Immunol.* **185**, 4777–4782 (2010).
109. Maul, R. W. *et al.* Uracil residues dependent on the deaminase AID in immunoglobulin gene variable and switch regions. *Nature Immunol.* **12**, 70–76 (2011).
110. Cho, R. J. *et al.* A genome-wide transcriptional analysis of the mitotic cell cycle. *Mol. Cell* **2**, 65–73 (1998).
111. Pybus, C. *et al.* Transcription-associated mutation in *Bacillus subtilis* cells under stress. *J. Bacteriol.* **192**, 3321–3328 (2010).
112. Maldonado, E. *et al.* A human RNA polymerase II complex associated with SRB and DNA-repair proteins. *Nature* **381**, 86–89 (1996).
113. Deem, A. *et al.* Break-induced replication is highly inaccurate. *PLoS Biol.* **9**, e1000594 (2011).
114. Hicks, W. M., Kim, M. & Haber, J. E. Increased mutagenesis and unique mutation signature associated with mitotic gene conversion. *Science* **329**, 82–85 (2010).
115. Strathern, J. N., Shafer, B. & McGill, C. B. DNA synthesis errors associated with double-strand-break repair. *Genetics* **140**, 965–972 (1995).
116. Gonzalez-Barrera, S., Garcia-Rubio, M. & Aguilera, A. Transcription and double-strand breaks induce similar mitotic recombination events in *Saccharomyces cerevisiae*. *Genetics* **162**, 603–614 (2002).
117. Saxe, D., Datta, A. & Jinks-Robertson, S. Stimulation of mitotic recombination events by high levels of RNA polymerase II transcription in yeast. *Mol. Cell. Biol.* **20**, 5404–5414 (2000).
118. Schildkraut, E., Miller, C. A. & Nickoloff, J. A. Transcription of a donor enhances its use during double-strand break-induced gene conversion in human cells. *Mol. Cell. Biol.* **26**, 3098–3105 (2006).
119. Derr, L. K. & Strathern, J. N. A role for reverse transcripts in gene conversion. *Nature* **361**, 170–173 (1993).
120. Fink, G. R. Pseudogenes in yeast? *Cell* **49**, 5–6 (1987).
121. Storici, F., Bebenek, K., Kunkel, T. A., Gordenin, D. A. & Resnick, M. A. RNA-templated DNA repair. *Nature* **447**, 338–341 (2007).
122. Sekiguchi, J. & Shuman, S. Site-specific ribonuclease activity of eukaryotic DNA topoisomerase I. *Mol. Cell* **1**, 89–97 (1997).
123. Henningfeld, K. A. & Hecht, S. M. A model for topoisomerase I-mediated insertions and deletions with duplex DNA substrates containing branches, nicks, and gaps. *Biochemistry* **34**, 6120–6129 (1995).
124. Maizels, N. Immunoglobulin gene diversity. *Annu. Rev. Genet.* **39**, 23–46 (2005).
125. Chiarle, R. *et al.* Genome-wide translocation sequencing reveals mechanisms of chromosome breaks and rearrangements in B cells. *Cell* **147**, 107–119 (2011).

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Competing interests statement

The authors declare no competing financial interests.

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